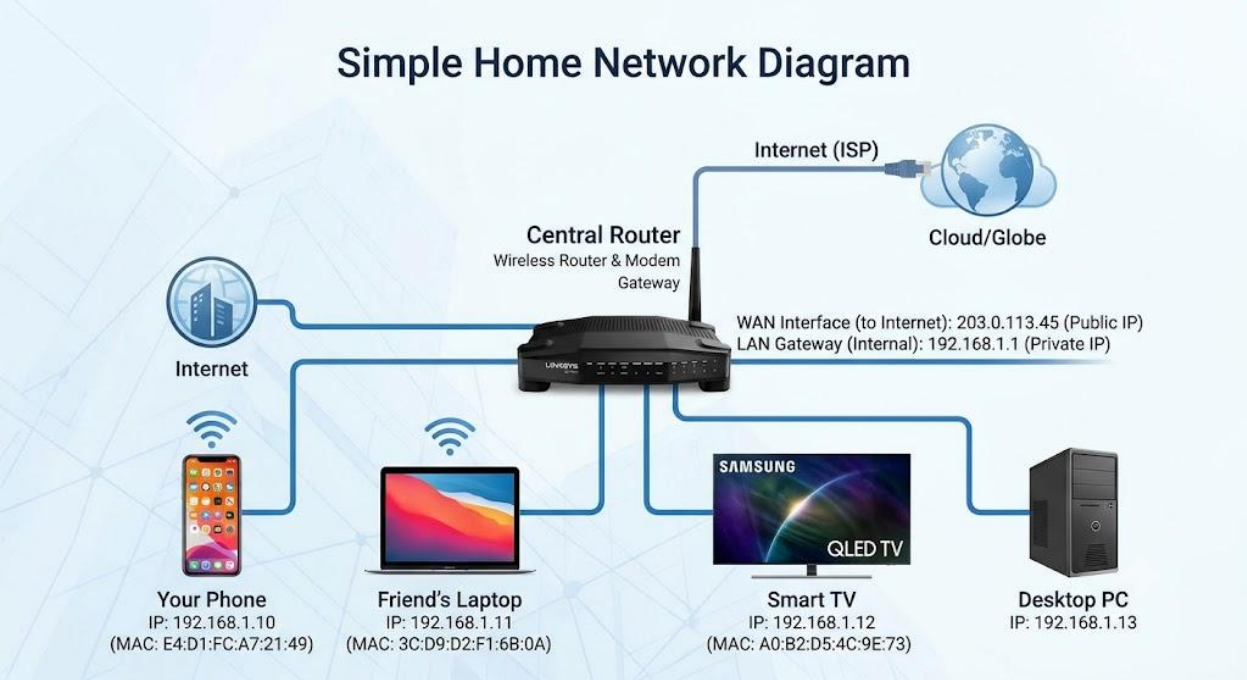
IP ROUTING & ROUTING PROTOCOLS

1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

   Ans:-



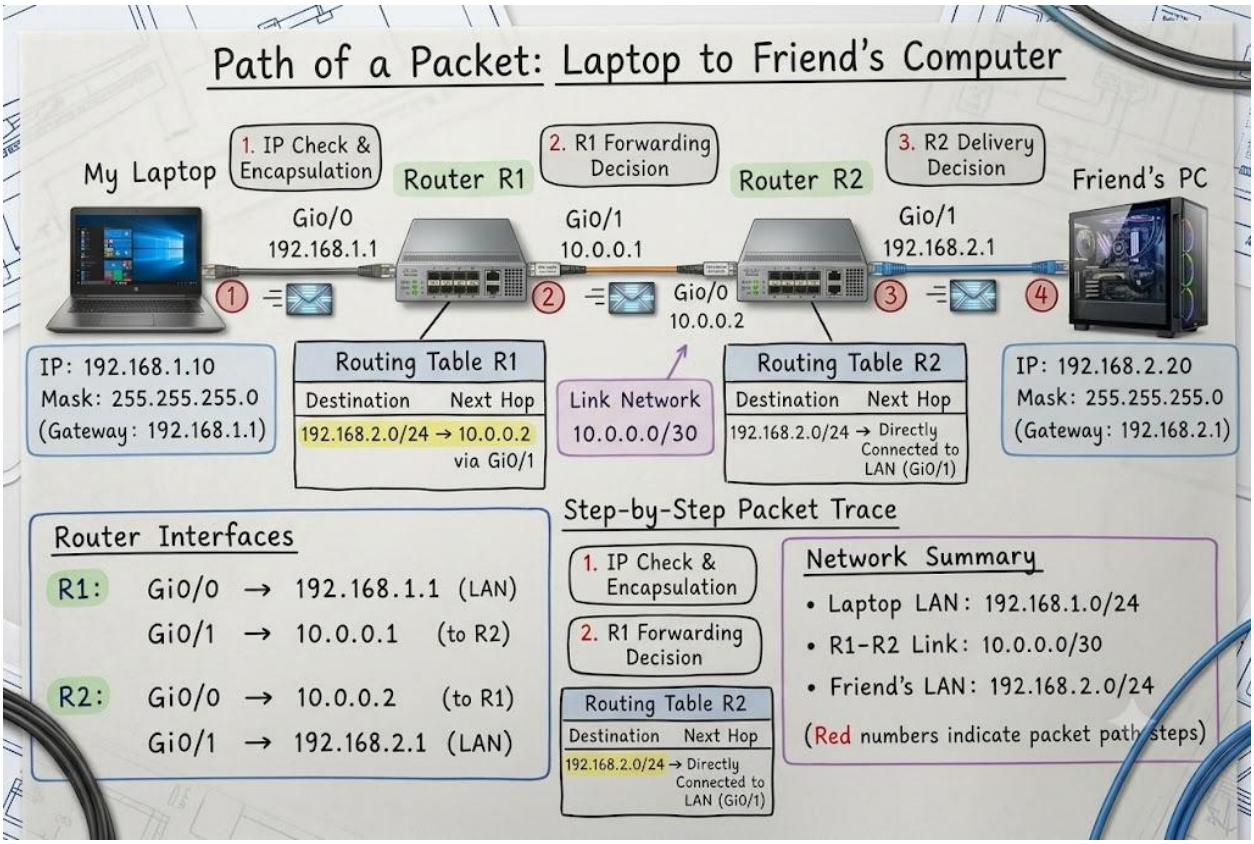
- **Phone:** 192.168.1.10/24, connected to router interface Gi0/0 (192.168.1.1).
- **Friend's laptop:** 192.168.2.10/24, connected to router interface Gi0/1 (192.168.2.1).
- **Smart TV:** 192.168.3.10/24, connected to router interface Gi0/2 (192.168.3.1).

Each device is in a different subnet, so the router is needed for them to talk to each other.

2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

Ans:-



Step 1 - My laptop: The laptop compares the destination with its own network. 192.168.2.20 AND 255.255.255.0 = 192.168.2.0, which is not 192.168.1.0, so the destination is not on my LAN. The laptop sends the packet to its default gateway, 192.168.1.1 (R1). Routing table checked:

Destination network	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	On-link (direct)	Ethernet / WiFi
0.0.0.0 (default)	0.0.0.0	192.168.1.1 (R1)	Ethernet / WiFi

Step 2 - Router R1: R1 receives the packet on Gi0/0 and reads the destination IP 192.168.2.20. It looks for the best (longest) match in its routing table. 192.168.2.0/24 matches, with next hop 10.0.0.2, so R1 reduces the TTL by 1 and forwards the packet out of Gi0/1 to R2. Routing table checked:

Destination network	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	Directly connected	Gi0/0
10.0.0.0	255.255.255.252	Directly connected	Gi0/1
192.168.2.0	255.255.255.0	10.0.0.2 (R2)	Gi0/1

Step 3 - Router R2: R2 receives the packet on Gi0/0 and checks its table. 192.168.2.0/24 is directly connected to Gi0/1, so no next router is needed. R2 reduces the TTL again and delivers the packet directly to 192.168.2.20 on its LAN. Routing table checked:

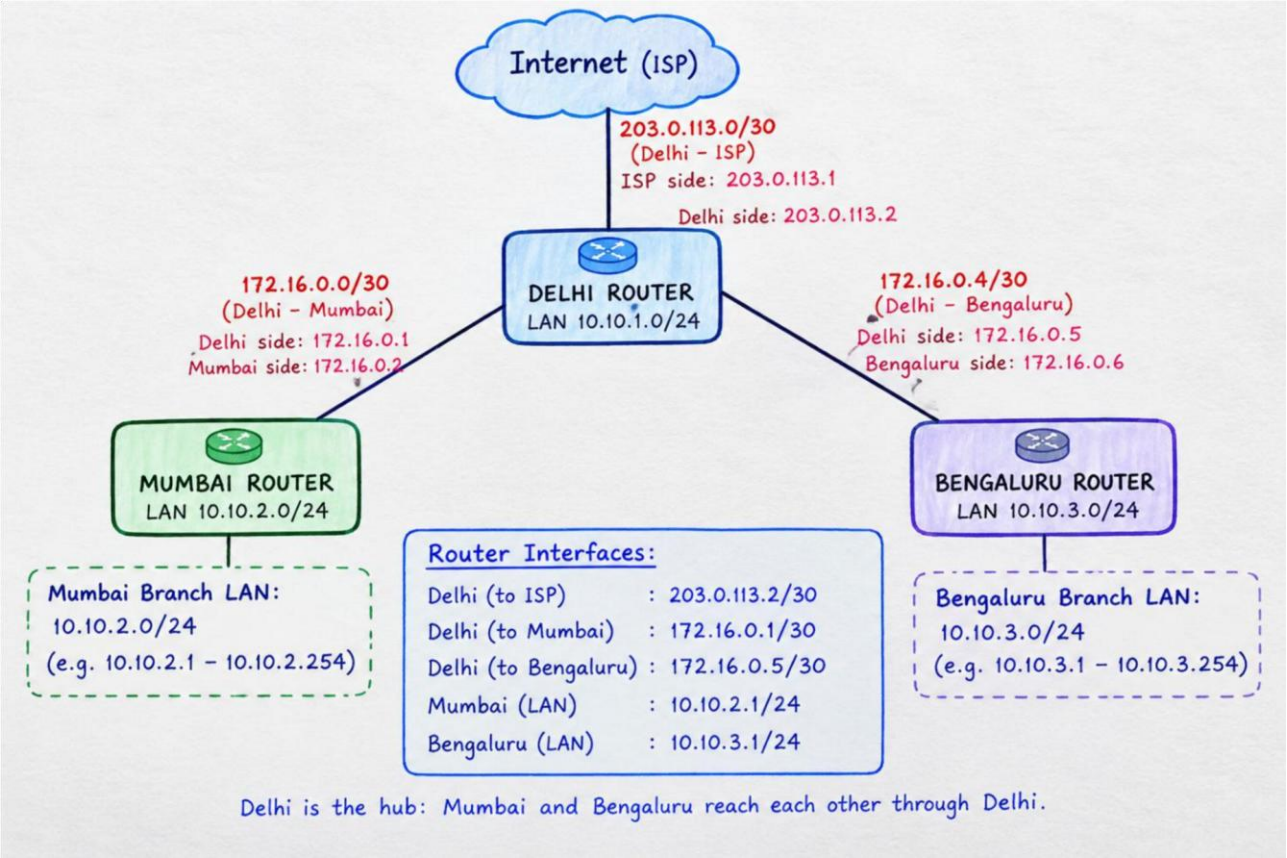
Destination network	Subnet mask	Next hop	Interface
10.0.0.0	255.255.255.252	Directly connected	Gi0/0
192.168.2.0	255.255.255.0	Directly connected	Gi0/1
Destination network	**Subnet mask**	**Next hop**	**Interface**
192.168.1.0	255.255.255.0	10.0.0.1 (R1)	Gi0/0

Step 4 - Friend's computer: The computer receives the packet. The reply travels back the same way: it uses its default gateway 192.168.2.1 (R2), and R2 sends it to 10.0.0.1 (R1) using the entry for 192.168.1.0/24, and R1 delivers it to my laptop.

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Ans:-

The three branches are Delhi (10.10.1.0/24), Mumbai (10.10.2.0/24) and Bengaluru (10.10.3.0/24). Delhi is the hub, and the cities are joined by small /30 links.



Routing table of the Delhi router:

Destination network	Subnet mask	Next hop	Interface
10.10.1.0 (Delhi LAN)	255.255.255.0	Directly connected	Gi0/0
10.10.2.0 (Mumbai LAN)	255.255.255.0	172.16.0.2	Se0/0 (link to Mumbai)
10.10.3.0 (Bengaluru LAN)	255.255.255.0	172.16.0.6	Se0/1 (link to Bengaluru)
172.16.0.0 (Delhi-Mumbai link)	255.255.255.252	Directly connected	Se0/0
Destination network	**Subnet mask**	**Next hop**	**Interface**
172.16.0.4 (Delhi-Bengaluru link)	255.255.255.252	Directly connected	Se0/1
0.0.0.0 (default route)	0.0.0.0	203.0.113.1 (ISP)	Gi0/2 (link to ISP)

Mumbai and Bengaluru routers have the same type of table: a connected route for their own LAN, and routes to the other cities with Delhi as the next hop.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Ans:-

Type	Pros	Cons	Real-world example
Static routing	1. Simple to set up on small networks 2. No extra CPU, memory or bandwidth used for route updates 3. Predictable path and more secure, since routes do not change on their own	1. Routes must be added by hand on every router 2. Does not adapt to a link failure, so traffic is lost until fixed 3. Hard to manage as the network grows	A small cafe or home router uses one static default route to its ISP to open Instagram and Zomato.
Dynamic routing (RIP / OSPF)	1. Learns routes automatically 2. Finds another path if a link fails 3. Scales to large networks and new branches	1. Uses CPU, memory and bandwidth for updates 2. More complex to configure and troubleshoot 3. A wrong or fake update can spread and cause problems	A food delivery app with servers in several cities: if the Delhi-Mumbai link fails, OSPF reroutes through Bengaluru so orders still go through.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Ans:-

AI tool used: Claude (AI assistant). **Prompt:** "Generate a sample configuration for OSPF routing between two routers."

The two routers R1 and R2 are joined by the link 10.0.0.0/30. R1 has the LAN 192.168.1.0/24 and R2 has the LAN 192.168.2.0/24 (same as Q2).

AI-generated configuration for R1 (Cisco IOS):

```
hostname R1
!
interface GigabitEthernet0/0  ip address
10.0.0.1 255.255.255.252  no shutdown
!
interface GigabitEthernet0/1  ip address
192.168.1.1 255.255.255.0  no shutdown
!
router ospf 1  router-id 1.1.1.1  network
10.0.0.0 0.0.0.3 area 0  network 192.168.1.0
0.0.0.255 area 0  passive-interface
GigabitEthernet0/1
! end write
memory
```

AI-generated configuration for R2:

```
hostname R2
!
interface GigabitEthernet0/0  ip address
10.0.0.2 255.255.255.252  no shutdown
!
interface GigabitEthernet0/1  ip address
192.168.2.1 255.255.255.0  no shutdown
!
router ospf 1  router-id 2.2.2.2  network
10.0.0.0 0.0.0.3 area 0  network 192.168.2.0
0.0.0.255 area 0  passive-interface
GigabitEthernet0/1
! end write
memory
```

Explanation of each line (R1; R2 is the same with its own addresses):

Configuration line	What it does
hostname R1	Sets the router's name to R1 so it is easy to identify.
interface GigabitEthernet0/0	Enters the settings of the port that connects to the other router.
ip address 10.0.0.1 255.255.255.252	Gives this port the IP 10.0.0.1 with a /30 mask (a small link network for two routers).
no shutdown	Turns the port on. Router ports are off by default.
interface GigabitEthernet0/1	Enters the settings of the port that connects to the LAN (laptops, phones).
ip address 192.168.1.1 255.255.255.0	Gives the LAN port the IP 192.168.1.1, which is the gateway for the LAN devices.
router ospf 1	Starts the OSPF routing process. 1 is the process ID, which only has meaning on this router.
router-id 1.1.1.1	Gives the router a unique name (ID) in OSPF so other routers can identify it.
Configuration line	**What it does**
network 10.0.0.0 0.0.0.3 area 0	Turns on OSPF on interfaces inside 10.0.0.0/30 and puts them in area 0 (the backbone area). 0.0.0.3 is the wildcard mask, the opposite of 255.255.255.252.
network 192.168.1.0 0.0.0.255 area 0	Advertises the LAN 192.168.1.0/24 to the other router. 0.0.0.255 is the wildcard mask for /24.
passive-interface GigabitEthernet0/1	Stops OSPF messages from being sent to the LAN (no routers there), but the LAN is still advertised. This is safer and saves bandwidth.
end / write memory	Leaves configuration mode and saves the settings so they stay after restart.

To check: show ip ospf neighbor (shows R2 as a neighbor) and show ip route ospf (shows the route to the other LAN